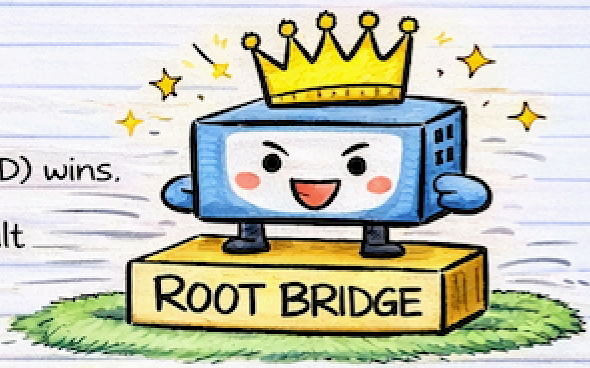
SPANNING TREE PROTOCOL

802.1D

802.1w

1. The Core Objective

Root Bridge: The "Master" switch.
The switch with the lowest Bridge ID (BID) wins.



- Bridge ID: Composed of Priority (default 32,768) + MAC Address.
- Modification: To force a specific switch to be the Root, manually lower its priority to 4096 or 0.

3. Port Roles & States

Role	Description	State Logic
Root (RP)	The "cheapest" path to the Root Bridge (1 per	Forwarding
Designated Port (DP)		200,000
Alternate/Backup		20,000

5. Port Transition (Classic STP):

- RSTP (802.1w): "Rapid" Uses a hanasrake= → Forwarding
- MSTP (802.1s): "Multiple" STP. Groups multiple VLANs into a single instance to save CPU resources

Visual Layout Reminders:

- Highlight (Yellow Neon): Root Bridge, RSTP, PortFast, BPDU Guard, Convergence.
- Circle (Red Pen): 802.1D, 802.1w, 802.1s.
- Doodles: A crown on the Root Bridge, a rocket, for RSTP, and shield

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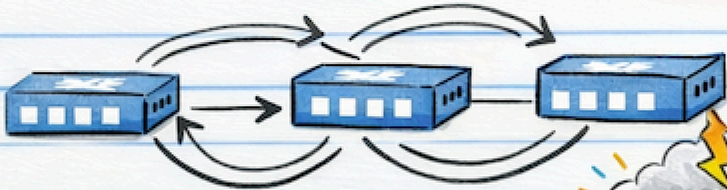
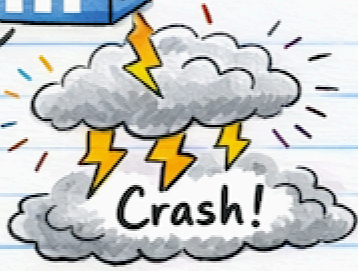
Spanning Tree Protocol 802.1D

A Layer 2 Protocol to prevent Switching Loops!

Without STP:

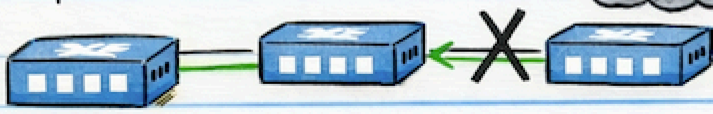
- Frames endlessly loop causing a "Broadcast Storm!"





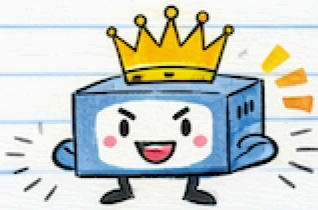
With STP:

- Keeps the network stable!



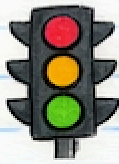
Key Topics:

- **The Election Process:**
  - Choosing the "Root Bridge" 👑 (The Boss!)



- **Port States & Roles:**
  - Blocking, Listening, Learning, Forwarding

Safety First!



STP Variations:

- Original STP, 802.1D
- Rapid STP (RSTP), 802.1w
- Multiple STP (MSTP) 802.1s

STP Keeps the Network Loop-Free!

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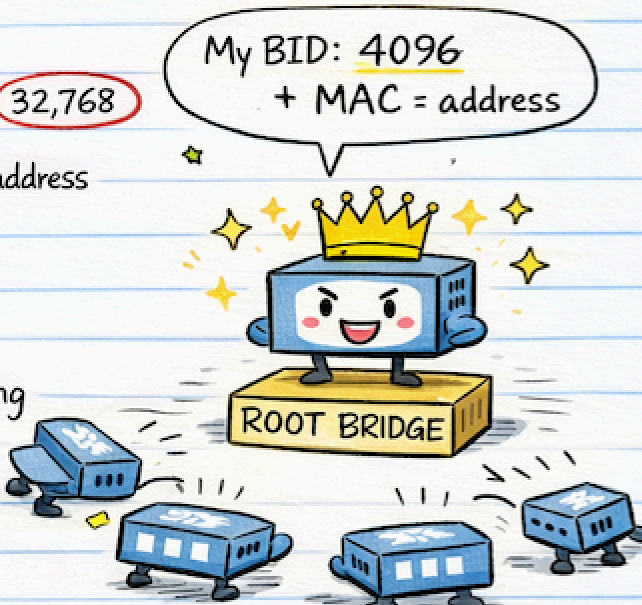
Section 1: The Root Bridge Election

In a network of switches, one switch must become the "Root Bridge." The switch with the lowest Bridge ID (BID) becomes the boss!

Bridge ID (BID):

- Bridge Priority: A number (default=32,768
- MAC Address: The unique hardware address of the switch

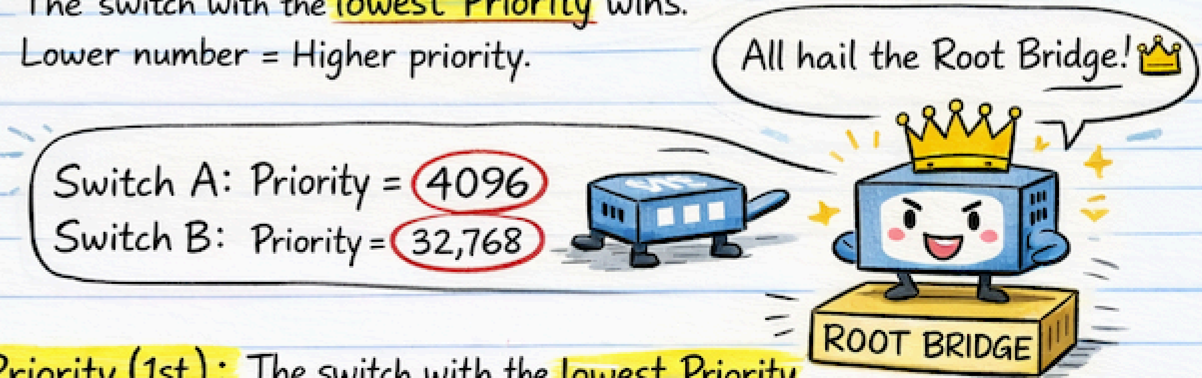


Example Story:

Imagine a room full of switches shouting their names (BIDs) The one with the lowest BID wins!

Priority (1st):

- The switch with the lowest Priority wins. Lower number = Higher priority.



Priority (1st): The switch with the lowest Priority

Lower number = Higher priority.

Lowest Bridge ID (Priority wins 1st!) → Becomes the Root Bridge!

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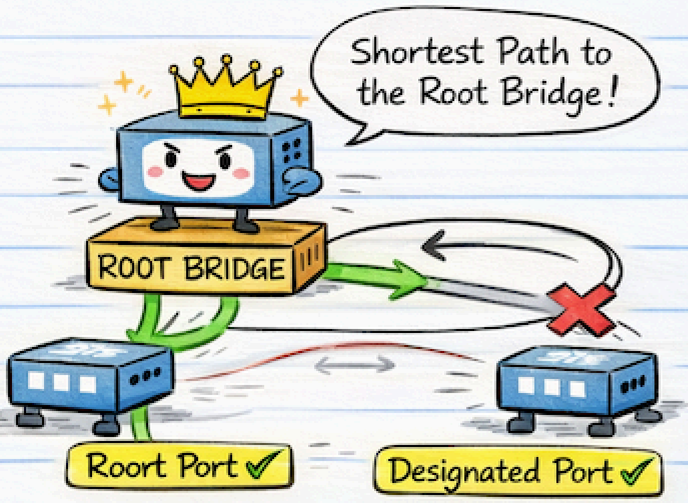
Section 2: Port Roles

Once the Root Bridge is elected, every other switch (Non-Root Bridges) must decide which of its ports will carry traffic and which will stay quiet to prevent loops.

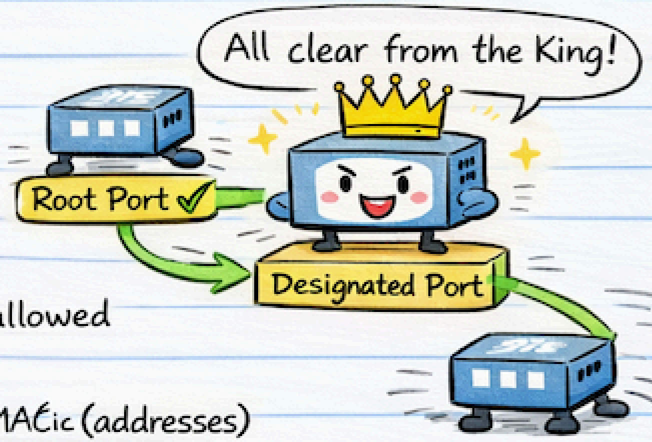


Root Port (RP):

- The port on a non-root switch that has the "cheapest" path to the Root Bridge.
  Each switch only gets one RP.

Designated Port (DP):

- The port on a segment (link) that is allowed to send traffic.
  The Root Bridge's ports are always Designated Ports.

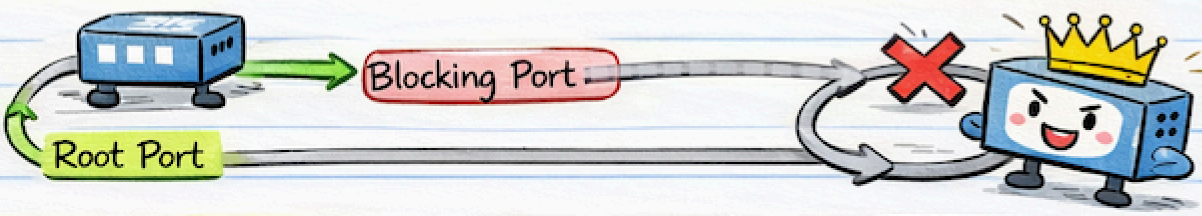


Blocking Port:

- Ports that are physically connected but "turned off" logically to break the loop.

- Blocking (Red light): No traffic allowed
- Listening (Yellow light): Preparing
- Learning (Yellow light): Building MACic (addresses)
- Forwarding (Green light): Full forwarding!

Root Port + Designated Port → Safe heading to Root Bridge! ✓



Can the Root Bridge have a Blocking Port!

Nope! The Root Bridge is the King 👑. All of its ports are Designated Ports to send data.

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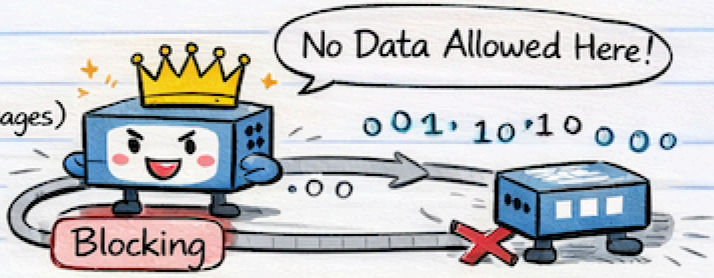
Section 3: Port States (The Logic Flow)

Ports don't just turn on instantly; they go through a "safety check" to make sure they won't cause a storm.

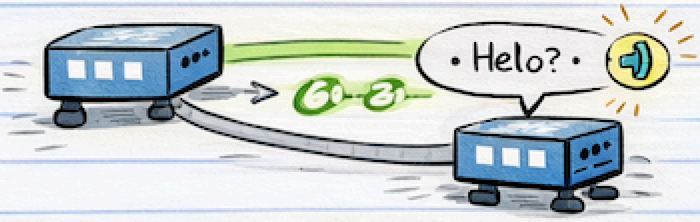
Blocking (Red light):

- Only receives BPDUs (STP messages) No data sent.



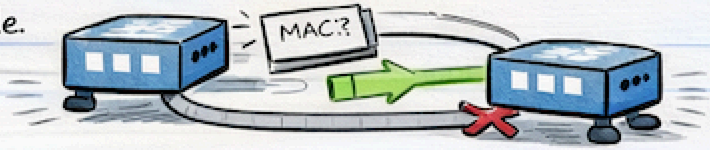
Listening (Yellow light):

- Checking for loops.



Learning (Yellow light):

- Building the MAC address table.

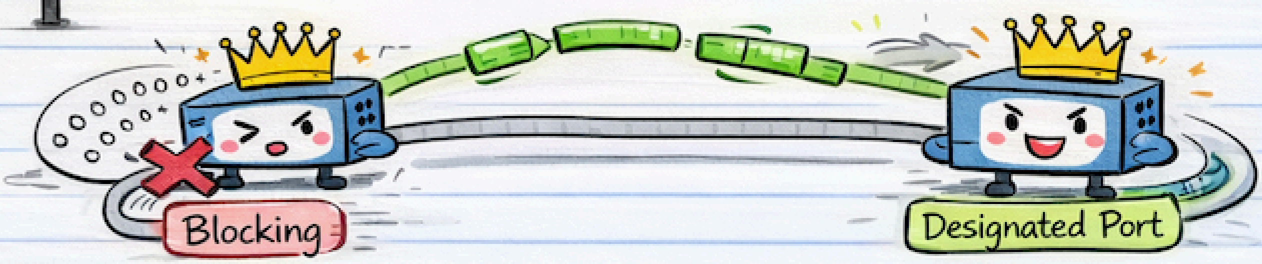


Forwarding (Green light):

- Data finally flows!

From Blocking ▶ Forwarding: Safety first!

Blocking (No Data) ➜ Listening (Checking) ➜ Learning (Building) ➜ Learning (Building) ➜ Forwarding (Full Flow!)



From Blocking ➜ Forwarding: Safety first!

Root Bridge is the King! All of its ports are Designated Ports to send data.

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Section 4: STP vs. RSTP (The Need for Speed)

Classic STP (802.1D) was reliable but slow, taking up to 50 seconds to recover from a link failure!

The IEEE developed Rapid Spanning Tree Protocol (RSTP) or 802.1w to fix the slow "Convergence" (the time it takes for the network to become stable again after a change).

Feature	Classic STP (802.1D)	Rapid STP (802.1w)
Speed	Slow (30-50 seconds)	Very Fast (under 2 seconds)
Port States	5 States (Blocking to Forwarding)	3 States (Discarding, Learning, Forwarding)
Port Roles	Root, Designated, Blocking	Root, Designated, Alternate, Backup

In RSTP, switches don't wait for timers! They converse!

Quick Knowledge Check:

Q: Unplugging a cable in Classic STP causes a 50-second network outage, even if there is a backup cable. Why?

A: Classic STP waits for timers to expire before reacting. That delay is why the whole network "freezes" for almost a minute after a failure, even when a backup link is available.

Rapid STP (RSTP) converges almost instantly!

RSTP uses "Proposal and Agreement" to skip the timers and start forwarding promptly!

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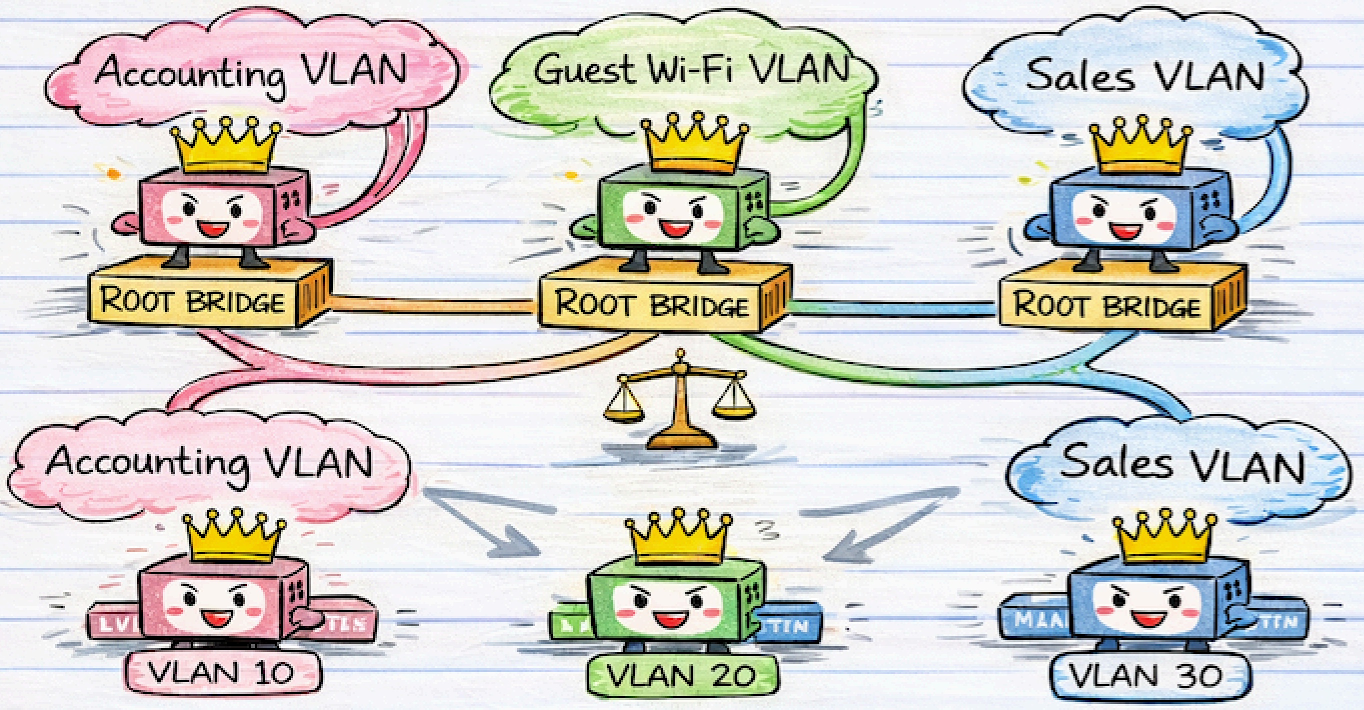
Section 5: STP Variations & VLANs

In a real-world network, you usually don't just have one big LAN. You have different VLANs (e.g., one for Accounting, one for Guest Wi-Fi).

Rapid STP (RSTP) replaces those timers with an active handshake . This allows you to have different Root Bridges for different VLANs, which helps balance the traffic load.



PVST+ (Per-VLAN Spanning Tree Plus): This is a Cisco-proprietary version. It runs a separate instance of STP for every VLAN.

This allows you to have different Root Bridges for different VLANs, which helps balance the traffic load.

MSTP (Multiple Spanning Tree Protocol - 802.1s):

Running a tree for every single VLAN can be hard on a switch's CPU.

MSTP maps multiple VLANs into a single "instance" or "group", reducing the workload while still allowing for Load Balancing.

Rapid STP (RSTP) converges almost instantly!

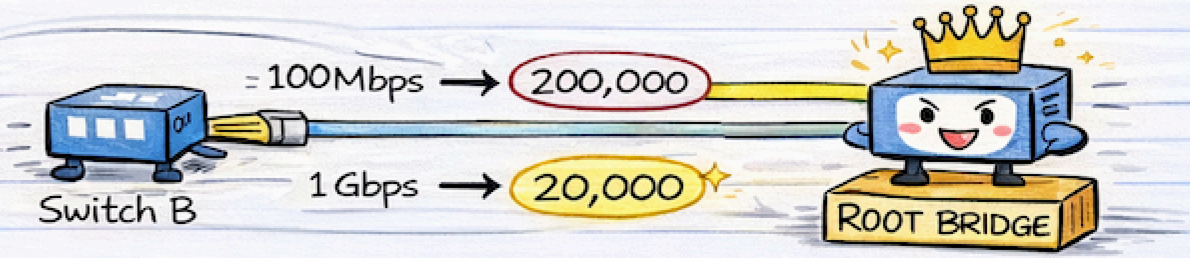
RSTP uses "Proposal and Agreement" to skip the timers and start forwarding promptly!

Path Costs

This is how switches calculate the "cheapest" (shortest) way to the Root Bridge. In your notes, these numbers help explain why one path is chosen over another.

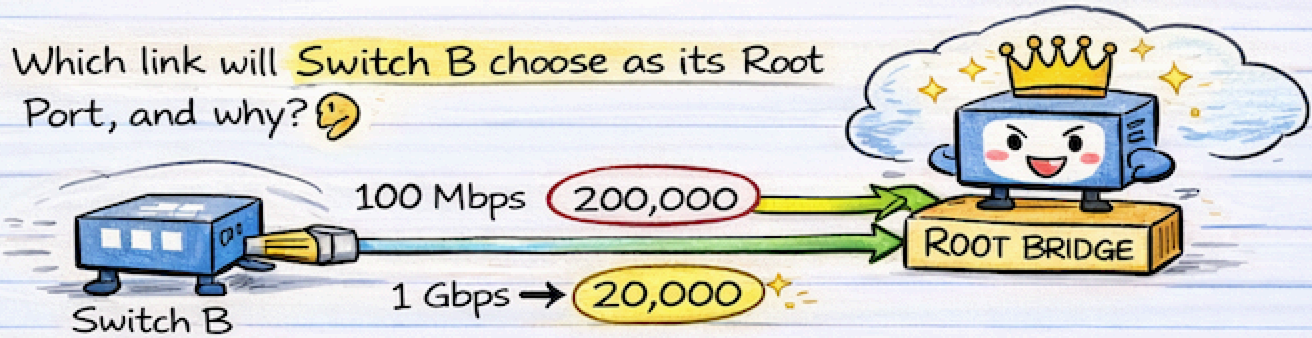
Link Speed	STP Cost (Classic)	STP Cost (Revised)
10 Mbps	100	2,000,000
100 Mbps	19	200,000
1 Gbps	4	20,000
10 Gbps	2	2,000

Imagine Switch B is connected to the Root Bridge by two cables. One is a 100 Mbps link and the other is a 1 Gbps link.



Imagine Switch B is connected to the Root Bridge by two cables.
One is a 100 Mbps link and the other is a 1 Gbps link.

Which link will Switch B choose as its Root Port, and why?



Since 1 Gbps has a lower cost (20,000)) than 100 Mbps (200,000), Switch B will choose the 1 Gbps link as its Root Port

In STP, the lower the cost, the better the path!

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STP Comprehensive Final Summary

1. The Core Objective

STP (802.1D) prevents **Layer 2 loops** in networks with redundant paths. It does this by logically blocking specific ports to create a loop-free tree topology. Without it, broadcast storms would crash the network.

2. The Election Process

Root Bridge: The **"Master"** switch.

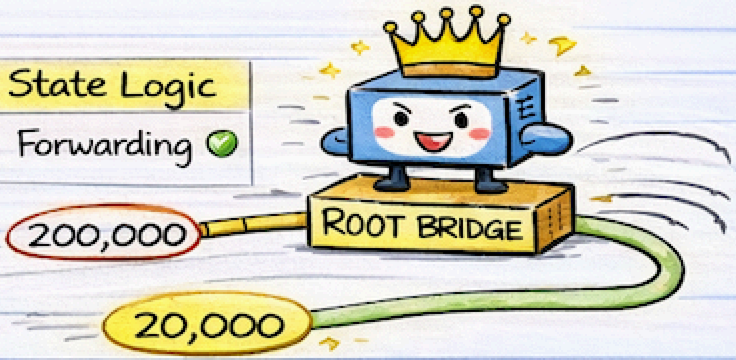
The switch with the **lowest Bridge ID** (BID) wins.

- **Bridge ID:** Composed of Priority (default 32,768) + MAC Address.
- **Modification:** To force a specific switch to be the Root, manually lower its priority to 4096 or 0.

State Logic		State Logic
Blocking	→	Listening
Listening	→	Learning
Forwarding	→	Forwarding
	→	Blocking

3. Port Roles & States

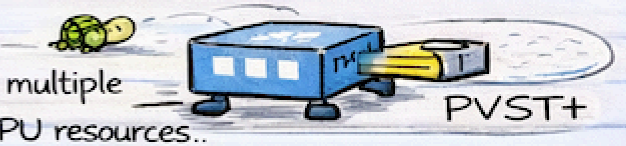
Role	Description	State Logic
Root Port (RP)	The "cheapest" path to the Root Bridge (1 per non-root switch)	Forwarding ✅
Designated Port (DP)		
Alternate/Backup		



Port Transition (Classic STP) Blocking ⛔ – Listening → Forwarding ✅

4. Protocol Evolution

- **RSTP** (802.1w): "Rapid" STP. Uses a handshake mechanism for sub-2-second convergence
- **MSTP** (802.1s): "Multiple" STP Groups multiple VLANs into a single instance to save CPU resources..



5. Path Cost & Speed:

Switches choose paths based on **Cost (lower is better)**:

10 Gbps : Cost 2,000

1 Gbps : Cost 20,000

100 Mbps: Cost 200,000

Doodles: A crown on the Root Bridge, a rocket for RSTP, and a shield

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